

James Mba Azam

Curriculum Vitae

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Professional Summary

Quantitative researcher and research software engineer with a PhD in Applied Mathematics, specialising in infectious disease modelling, outbreak analytics, and global health communication. Experienced in cross-institutional collaborations, teaching, mentoring, and developing open-source tools for real-time decision support and capacity building.

Research Interests

- Utilising mathematical and statistical models and methods to quantify the impact of outbreak response interventions to inform public health decision-making.
- Developing open-source software/tools and pipelines for reproducible outbreak response analytics.
- Investigating and contributing insights in effective science and public health policy communication.

Education

Feb, 2018 – Apr, 2022 **PhD Applied Mathematics**, *DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA)*, Stellenbosch University, South Africa

Thesis title *Modelling outbreak response intervention strategies for decision-making*

Supervisors Prof. Juliet R.C. Pulliam & Prof. Matthew J. Ferrari

Mar, 2016 – Mar, 2018 **MSc Mathematics**, *DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA)*, Stellenbosch University, South Africa

Thesis title *Assessment of a measles outbreak response vaccination campaign, and two measles parameter estimation methods*

Supervisor Prof. Juliet R.C. Pulliam

Aug, 2014 – Jun, 2015 **Structured MSc Mathematical Sciences**, *African Institute for Mathematical Sciences (AIMS)*, Senegal

Thesis title *Hedging Conditional Value at Risk*

Supervisor Dr. Maciej J. Capinski

Aug, 2009 – Jun, 2013 **BSc Actuarial Science**, *Kwame Nkrumah University of Science and Technology*, Ghana

Thesis Title *Predicting optimal periods for the sale of cocoa in Ghana for the year 2013, using futures contracts*

Supervisor Mr. Vincent K. Dedu

Professional Experience

- Sept, 2025 – Present **Research Fellow**, *Centre for Mathematical Modelling of Infectious Diseases, London School of Hygiene & Tropical Medicine*, London, United Kingdom
- Duties:**
- Developing spatial models to predict the risk of polio re-emergence and analyse eradication strategies.
 - Analysing multi-pathogen wastewater surveillance data as part of the South African Wastewater Consortium in collaboration with the National Institute for Communicable Diseases (NICD), the South African Centre for Epidemiological Modelling and Analysis (SACEMA), the Modelling and Simulation Hub, Africa (MASHA), and the Gauteng City-Region Observatory (GCRO).
- Nov, 2022 – Aug, 2025 **Research Software Engineer**, *Epiverse Initiative, London School of Hygiene & Tropical Medicine*, London, United Kingdom
- Duties:**
- Lead and contribute to the development and maintenance of open-source R packages for outbreak analytics, including EpiNow2, epichains, and bpmmodels.
 - Conduct infectious disease modelling research, including the planning, analysis, and interpretation of analyses for outbreak response and preparedness.
 - Develop technical and non-technical resources such as R package vignettes, function references, tutorials, how-to guides, and blog posts.
 - Provide and support outbreak analytics and data science trainings through Epiverse-TRACE.
 - Support outbreak analytics open-source communities, including epiforecasts and epinowcast, through code contributions, code reviews, seminars, and training activities.
- Mar, 2025 – Jul, 2025 **Technical Reviewer and Video Content Creator**, *Introduction to Infectious Disease Modelling for Public Health*, Epiverse TRACE Initiative Online Course
- Reviewed and edited English translations of outbreak analytics training materials originally developed in Spanish, ensuring clarity, technical accuracy, and alignment with the African context, including using examples from Africa.
 - Created English versions of video explainers on modelling topics such as SIR models, Zika compartmental modelling, and model simulation in R, originally created in Spanish.
- Feb, 2022 – Nov, 2022 **Junior Researcher**, *DSI-NRF Center of Excellence in Epidemiological Modelling and Analysis*, Stellenbosch, South Africa
- Duties:**
- Co-developed a mathematical model of polio to provide analytics decision support to South Africa's National Advisory Group on Immunization in preparation towards a potential threat of polio re-emergence.
 - Co-developed an analysis pipeline (using *R* and *GNU make*) for detecting changes in the trends of reported COVID-19 severity in South Africa.
 - Developed a model to analyse strategies for mitigating a hypothetical pathogen in the face of emerging variants.
 - Contributed to the activities of SACEMA Modelling and Analytics Response Team (SMART).

- Feb, 2021 – Sep, 2021 **Data Science Consultant (Part-time)**, *Winning Insights and Models for Marvellous Yields (Wimmy)*, Cape Town, South Africa
- Duties:**
- Extended an existing COVID-19 model to analyse the impact of vaccine rollout in South Africa, Chile, Peru, and Zimbabwe.
 - Contributed to a process to improve the calibration, validation, and run-time of the COVID-19 model.
 - Led and contributed to LinkedIn articles about various health issues of immediate concern, including the potential impact of emerging COVID-19 variants, and addressing the issue of blood clotting following COVID-19 vaccination.
 - Drafted guidelines for code testing and review within Wimmy.
 - Contributed to the design of proposals for new analyses for Anglo America, including a study on Noise-Induced Hearing Loss among miners.

Research Affiliations

- Jan, 2023 – Present **Research Associate**, *South African Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA)*, Stellenbosch University, South Africa
Honorary affiliation in recognition of sustained contributions to SACEMA's research and training activities. The role fosters collaborative research, training, and data sharing.
- Jan, 2025 – Present **Senior Research Fellow**, *The German West Africa Centre for Global Health and Pandemic Prevention (G-WAC)*, Kwame Nkrumah University of Science and Technology, Ghana
Honorary appointment in recognition of contributions to GWAC's regional capacity building initiatives in infectious disease modelling and global health research. This role is aimed at facilitating collaborative research, training, and data sharing.

Fellowships, Honors, and Awards

- Sept, 2020 & Sept, 2021 **Best student oral presentation**, *DSI-NRF SACEMA Research Days*
- Dec, 2019 **Stellenbosch University Library Open Access Fund, £405**
A competitively awarded subsidy towards an Open Access Publication fee.
- May, 2019 **Professional Certificate Program in Health and Humanitarian Supply Chain Management, \$5,000/6,000 of course fee waived**, Georgia Tech Center for Health & Humanitarian Systems, Georgia, United States
A course fee waiver awarded competitively. I could not attend because of unforeseen circumstances.
- Oct, 2018 **Big Data Policy Fellowship, £4000**, Development in Africa with Radio Astronomy (DARA), University of Manchester, United Kingdom
A one month fellowship that provided training in science and policy communication in the context of Big Data projects and in particular, the Square Kilometre Array in South Africa and Australia.
- Feb, 2018 **PhD Bursary, R 140, 000**, DSI-NRF South African Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University
South Africa
- Aug, 2017 **Best student oral presentation**, *DSI-NRF SACEMA Research Days*
- Feb, 2016 **Masters Bursary, R 120, 000**, DSI-NRF SACEMA, Stellenbosch University
South Africa

Presentations

Conference Oral Presentations

- Apr, 2025 epichains: an R package for analysing transmission chains data. *Humanitarian Modelling Dialogue Workshop, WHO Pandemic Hub, Berlin, Germany*
- Nov, 2024 Benchmarking time-varying reproduction number, R_t , and forecasting models - a case study of the EpiNow2 R package. *Infectious Disease Modelling (IDM) Conference, Bangkok, Thailand*
- Aug, 2023 Superseding R packages – lessons learned from transitioning to the epichains R package from bmodels. *Research Software London Southeast conference, Imperial College, London, United Kingdom*
- Nov, 2021 **JM Azam**, MJ Ferrari, JRC Pulliam. Modelling the relative benefits of using the measles vaccine outside cold chain for outbreak response. *Epidemics8, Elsevier (Virtual presentation)*
- Oct, 2019 **JM Azam**, MJ Ferrari, JRC Pulliam. Measles outbreak response: is it worthwhile to use the vaccine outside of the cold chain?. *Modelling in the Context of African Health (MoCAH), Durban, South Africa.*
- Sept, 2019 **JM Azam**, P Xiaoxi, AB Elisha, MJ Ferrari, JRC Pulliam. Outbreak response models of human vaccine preventable diseases (1970-2019) - A systematic review. *DSI-NRF SACEMA Research Days (PhD Student Presentation), Stellenbosch, South Africa.*
- Sept, 2017 **JM Azam**, JRC Pulliam. Measles in a school in Stellenbosch, South Africa. *DSI-NRF SACEMA Research Days (MSc Student Presentation), Stellenbosch, South Africa.*
- Invited Talks
- Nov, 2024 Epiverse-TRACE: Advancing and Enabling Collaborative Open-Source R Tools for Outbreak Analytics, JUNIPER Weekly Seminar (virtual)
- Nov, 2024 Summary of analytical and statistical tools available for modelling pandemic H5N1 containment and mitigation, WHO R&DBlueprint Virtual Consultation
- Sept, 2024 Mpox 2024 outbreak: Common analytics tasks and available R tools/packages, UK Pandemic Sciences Network
- Sept, 2024 Mpox 2024 outbreak: Common analytics tasks and available R tools/packages, World Health Organisation Mpox Analytics Collaboratory Seminars
- Jun, 2022 *Modelling outbreak response strategies for decision-making (Presented virtually), Center for Epidemiological Modelling and Analysis, University of Nairobi, Kenya.*
- Jul, 2020 *Modelling the benefits of using the measles vaccine outside cold chain for outbreak response, Médecins Sans Frontières (MSF) Vaccine Working Group meeting (Presented virtually)*
- Apr, 2019 *Measles outbreak response vaccination: the role of the vaccine supply chain in low-income settings, Department of Logistics, Stellenbosch University*
- Poster Presentations
- Dec, 2019 **JM Azam**, MJ Ferrari, JRC Pulliam. A framework for evaluating vaccine delivery strategies for responding to measles outbreaks in hard-to-reach populations. *Epidemics7, Charleston, United States (Presented on my behalf).*
- Sept, 2018 **JM Azam**, JRC Pulliam. Measles outbreak response vaccination: how can we optimize the vaccine supply chain for efficient responses?. *DSI-NRF SACEMA Research Days (PhD Student Presentation), Stellenbosch, South Africa.*
- Jun, 2016 **JM Azam**, MI Andersson, E du Toit, M Nieuwoudt. Modelling Hepatitis B Transmission in sub-Saharan Africa - A Focus on the Impact of Vaccination Within Existing HIV MTCT Infrastructure. *Meaningful Modelling of Epidemiological Data, MMED (Student Presentation), African Institute for Mathematical Sciences (AIMS), Muizenberg, South Africa.*

Panel Discussions

- Nov, 2024 **WHO R&DBlueprint Virtual Consultation.** *Preparing for containment and mitigation of pandemic H5N1 influenza.* We discussed gaps in our knowledge and Present approach to tackling a potential H5N1 widespread outbreak.
- Jul, 2023 **Health Data Research UK Black Internship Programme.** *Career Panel.* We shared our career journeys and experiences, and answered questions from the mentees.
- Dec, 2020 **The New African Scientist.** *STEM - For Us, By Us conference, Visibility STEM Africa.* We discussed on topics including the future of the academic/research environment in Science Technology Engineering Mathematics (STEM) in Africa, mental health among young African researchers, how we find the motivation to do work outside of research, and our vision for STEM in Africa in the next few years.
- Dec, 2018 **Science Communication for Policy Engagement.** *Science Forum South Africa (SFSa), CSIR International Convention Centre, Pretoria, South Africa.* We discussed the relationship between science communication and science engagement, and the challenges that early career researchers face in communicating the policy implications of their research. We also discussed (South) Africa's readiness for the Fourth Industrial Revolution and the emerging challenges.

Teaching Experience

- Sept, 2024 – Current **Lead facilitator,** *Modelling for Pandemic Preparedness and Response*, German West African Center for Global Health and Pandemic Prevention, Kwame Nkrumah University of Science and Technology, Ghana
- June, 2025 **Co-lead facilitator,** *Outbreak Data Science for Lab Scientists: A Hands-On Introduction with the EpiVerse R Package Ecosystem*, KEMRI Wellcome Trust Research Programme, Kilifi, Kenya
- Aug, 2022 **Co-lead facilitator,** *Introduction to R: Management, Exploration, and Communication of Data*, SACEMA, South Africa
- Oct, 2017; May, 2018 & 2019 **Tutor (assistant),** *Introduction to R: Management, Exploration, and Communication of Data*, SACEMA, South Africa
- Aug, 2016 – Jun, 2017 **Teaching assistant,** *Introductory Calculus and Algebra*, Department of Mathematical Sciences, Stellenbosch University, South Africa
- Sept, 2013 – Jul, 2014 **Teaching assistant,** *Introductory Probability and Statistics, and R for Statistical Analyses*, Department of Mathematics, Kwame Nkrumah University of Science and Technology, Ghana

Student Supervision and Mentoring

- Apr, 2026 – Apr, 2027 **Mentor,** *Measles Analytics Hub Mentorship Programme*, Imperial College, United Kingdom
I am mentoring and supervising a mentee based in Uganda to develop a statistical model for estimating subnational measles immunity profiles to support the Ugandan National Expanded Programme on Immunization (UNEPI).
- Jul, 2025 – Aug, 2025 **Supervisor,** *Health Data Research UK Black Internship Programme*, London School of Hygiene & Tropical Medicine, United Kingdom
I am supervising an intern over an 8-week period to develop applied skills in infectious disease forecasting and evaluation through a project assessing the performance of COVID-19 forecasts in the UK using R, EpiNow2, and scoringutils.
- Jul, 2024 – Aug, 2024 **Mentor,** *Health Data Research UK Black Internship Programme*, London School of Hygiene & Tropical Medicine, London, United Kingdom
For 8 weeks, I mentored an intern by providing advice on career trajectories, how to effectively conducting his research, and personal development in preparation for PhD studies.

Apr, 2023 – Jun, 2023 **Mentor**, *Research Software Camp*, The Software Sustainability Institute , United Kingdom
In 3 months, I mentored a PhD student to develop a working knowledge of using R for running generalised linear mixed models. We worked our way up from no knowledge of R to running and interpreting regression models.

Professional Development

Workshops and Trainings

- Dec, 2025 – Feb, 2026 **Diversifying Leadership Programme**, *Advance HE*, London
A 3-month leadership course that equipped me with a mindset to navigate institutional power dynamics by leveraging cultural capital and strategic networking. Developed practical skills in using Action Learning Sets to solve complex leadership challenges and refined a personal leadership story for increased visibility.
- Apr, 2025 **Humanitarian Modelling Dialogue Workshop**, *WHO Pandemic Hub*, Berlin, Germany
I co-facilitated a hackathon on software tools for rapid outbreak analytics during humanitarian crises.
- Feb, 2025 **UK Epidemic Analytics and Modelling Meeting**, *London School of Hygiene & Tropical Medicine (HPRU in Health Analytics and Modelling)*, London, United Kingdom
This was a 2-day meeting of modellers from across the UK to network and brainstorm the way forward with pro-active preparedness towards a more efficient epidemic analytics in the UK.
- Nov, 2024 **Understanding, evaluating, and improving forecasts of infectious disease burden**, *Infectious Disease Modelling Conference*, Bangkok, Thailand
This was a half-day pre-conference workshop.
- Apr, 2024 **R Developer Day**, *Imperial College*, London, United Kingdom
An open, collaborative event for people interested in contributing to base R. I worked with two others to submit a patch for an improvement to the base package's `substr()` and `substring()` functions to help clarify their similarities and differences and demonstrate some previously obscure features.
- Dec, 2023 **Curriculum development workshop for the short course, "Modelling for Pandemic Preparedness and Response"**, *German-West African Centre for Global Health and Pandemic Prevention/Technische Universität Berlin*, Berlin, Germany
This workshop involved a meeting of experts to design the named two week short course to be offered in Ghana.
- Jul, 2023 **RSLondonSouthEast 2023**, *Imperial College London*, London, United Kingdom
This workshop aimed to provide a regional focus on research software activities, offer an opportunity to present work to a regional audience, and help participants develop links with the wider Research Software Engineering community.
- Mar, 2023 **Forecasting natural and social systems**, *The Royal Society*, London, United Kingdom
This meeting aimed to convene leading experts in forecasting across disciplines to discuss the predictability of different systems, to exchange expertise, and to identify common challenges and possible solutions.
- Dec, 2022 **100 days and 100 lines of code**, *Epiverse Initiative*, *London School of Hygiene & Tropical Medicine*, London, United Kingdom
- Oct, 2022 **Measles modelling workshop**, *Vaccine Impact Modelling Consortium*, Virtual
- Nov, 2020 **FameLab - Science Communication Competition**, *DSI-NRF Center of Excellence in Epidemiological Modelling and Analysis*, Stellenbosch, South Africa
- Jan, 2020 **Software Engineering for Applied Mathematical Sciences**, *SACEMA*, Stellenbosch
- Nov, 2019 **The Annual African Vaccinology Course**, *Vaccines for Africa Initiative (VACFA)*, University of Cape Town, South Africa

- Feb, 2019 **FameLab - Science Communication Competition**, *DSI-NRF Center of Excellence in Epidemiological Modelling and Analysis*, Stellenbosch, South Africa
- May, 2018 **The Conversation Writing Workshop**, *Stellenbosch University*, South Africa
- Dec, 2017 **FameLab - Science Communication Competition**, *CSIR International Convention Centre*, Pretoria, South Africa
- Dec, 2017 **Spatial Disease Mapping**, *University of Cape Town*, Cape Town
- Nov, 2017 **The Use of Measles Vaccines Under Extended Controlled Temperature Conditions**, *Medicins sans Frontieres*, Paris, France
- May, 2017 **Data Science**, *AIMS-South Africa*, Muizenberg, Cape Town, South Africa
- Aug, 2016 **Science Communication and Public Engagement**, *AIMS-South Africa*, Muizenberg, Cape Town, South Africa
- Jul, 2016 **Clinic on the Meaningful Modeling of Epidemiological Data (MMED)**, *University of Florida, SACEMA-Stellenbosch University, AIMS-South Africa*, Muizenberg, Cape Town, South Africa
- May, 2016 **Global Change Impact on Infectious Diseases and Alien Species Expansion**, *AIMS-South Africa in collaboration with 16 other societies/organisations*
- Jul, 2015 **Computing in Applied Mathematical Sciences (CAMS)**, *AIMS-Ghana*
Short Courses
- Sept, 2023 **Equality & Diversity Essentials**, *London School of Hygiene & Tropical Medicine*, London
- Nov, 2022 **GDPR Compliance: Essential Training**, *LinkedIn Learning*
- Nov, 2020 **Learn to write for the media**, *The Conversation Africa*
An online course that teaches the basics of how to write media/popular articles based on scientific findings.
- Dec, 2019 **Science Communication Master Class**, *Stellenbosch University*, Stellenbosch
- Jul, 2018 **The Systematic Review**, *African Doctoral Academy*, Stellenbosch University, South Africa
- May, 2018 **Individual-based Modelling in Epidemiology - A Practical Introduction**, *SACEMA, Stellenbosch University*, South Africa
- August, 2017 **Introduction to Scientific Writing Skills**, *The Writing Center*, Stellenbosch University, South Africa
- Oct, 2016 **Bayesian Analysis of Longitudinal Data**, *SACEMA, Stellenbosch University*, South Africa
- Sept, 2016 **Mathematical Modeling of Infectious Diseases (MMID)**, *University of Cape Town*, Cape Town, South Africa
- Mar - Aug, 2016 **Introduction to Epidemiology and Research Methodology**, *SACEMA, Stellenbosch University*, South Africa
- May, 2016 **Applying Quantitative Bias Analysis to Epidemiologic Data**, *SACEMA, Stellenbosch University*, South Africa
- March, 2016 **Bayesian Biostatistics**, *SACEMA, Stellenbosch University*, South Africa

Peer Review

- Scientific Reports
- PLOS Global Health
- Journey of Open Science Software
- Bulletin of Mathematical Biology

Open Source Contributions

- Nov 2022 – Present **Developer & maintainer**, *epichains* & *bpmmodels*
R packages for simulating and analysing transmission chains using branching process models.
- Nov 2022 – Present **Co-maintainer**, *EpiNow2*
R package for estimating real-time infectious disease cases and time-varying epidemiological parameters.
- 2023 – Present **Contributor**, *epinowcast*, *primarycensored*, and *scoringutils*
R packages for forecasting and nowcasting real-time infectious disease cases and time-varying epidemiological parameters, and evaluation of forecasts.

Technical Competencies

- Programming tools R, Stan, Bash, GNU Make, Github & Git
- Dashboard tools RShiny, Quarto Dashboards
- Markup languages LaTeX, Markdown, R Markdown, HTML
- Operating Systems MacOS, Linux/Ubuntu, Microsoft Windows
- Presentation Quarto Presentations, MS PowerPoint, LaTeX Beamer
- Project management Agile/scrum
- Graphic Design Inkscape, Adobe suite, Camtasia, Canva, CapCut

Positions of Responsibility

- Dec, 2025 – Present **Elected Chair**, *Black Early Career Staff Network*, London School of Hygiene & Tropical Medicine, London
United Kingdom
- Apr, 2025 – Dec, 2025 **Acting Chair**, *Black Early Career Staff Network*, London School of Hygiene & Tropical Medicine, London
United Kingdom
- Jul, 2020 – Jul, 2021 **President**, *Association of Ghanaian Students in Stellenbosch University*, Stellenbosch University, South Africa
- Jun, 2019 – Jun, 2020 **Vice President**, *Association of Ghanaian Students in Stellenbosch University*, Stellenbosch University, South Africa
- 2017 **Student Representative**, *SACEMA Research Days planning committee*, Stellenbosch University, South Africa
- 2017 **Student Representative**, *SACEMA Strategic Planning Meeting*, Stellenbosch University, Stellenbosch University
- 2009 – 2013 **Member of Several committees**, *Pax Romana Local and Pax Romana Choir*, Kwame Nkrumah University of Science and Technology, Ghana

Diversity, Equity, and Inclusion

- Jan, 2025 – Present **Lead and Acting Chair**, *Black Early Career Staff Network Formation Task Force*, London School of Hygiene & Tropical Medicine, London, United Kingdom
- Dec, 2022 – Oct 2023 **Active Member**, *Equity, Diversity, Inclusion, and Accessibility Working Group*, Society of Research Software Engineering, United Kingdom
- Jun, 2023 – Sept, 2023 **Career Panel Member**, *Health Data Research UK Black Internship Programme*, London School of Hygiene & Tropical Medicine, United Kingdom

Professional Memberships

- Sept, 2024 – Present **Affiliate Member**, *Joint UNiversities Pandemic and Epidemiological Research (JUNIPER)*, United Kingdom

Dec, 2022 – Present **Member**, *Society of Research Software Engineering*, United Kingdom

Sep, 2023 – Present **Member**, *Models of Infectious Disease Agent Study (MIDAS)*, University of Pittsburgh
US

Referees

Available upon request.